



Towards Robust Uzbek Neural Dependency Parsing

Cross-Treebank Training & Linguistically-Motivated Subword Fusion

Sanatbek Matlatipov

National University of Uzbekistan named after Mirzo Ulugbek
`s.matlatipov@nuu.uz`

NLDB 2026 | 18 June 2026



Outline

Introduction

Related Work

Dataset

Methodology

Results & Discussion

Conclusion

Uzbek Language I

- ▶ **Growing NLP interest**, yet still under-resourced compared to major world languages.
- ▶ **Low-Resource**: NLP resources are sparse and low density.
- ▶ **Our language is Uzbek**:
 - ▶ Official language of Uzbekistan.
 - ▶ Native to Central Asia, Afghanistan, Russia, and Xinjiang (China).
 - ▶ **Spoken by 40 million** people (as of 2026).

Uzbek Language I

Introduction

Uzbek Language I

Research/Initial NLP

Progress

Shared tasks & Annotation

Evaluation & Conclusion

Speech Recognition

Uzbek Agent-based Sentiment Analysis for the Uzbek Language

- Not enough work done
- Low-Resource(NLP resources) ≠ low density
- Our Language is Uzbek
 - Official language of Uzbekistan
 - Native to all CA Countries, Afghanistan, Russia, Xinjiang(China)
 - Spoken by 37 million (as of 2024)

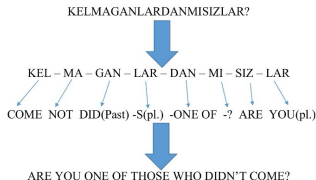



Why is Uzbek dependency parsing hard?

- ▶ **Agglutulative:** long suffix chains encode case, tense, person, evidentiality.
- ▶ **Flexible word order** and strong genre variation.
- ▶ **Low-resource:** fewer than 1,000 manually annotated UD sentences.

Turkic features

- ▶ Null subject (pro-drop).
- ▶ No articles or grammatical gender.
- ▶ SOV word order.



One Uzbek word = a whole English sentence — agglutination, plus scarce data, defines our two challenges.

Suffixes carry the syntax

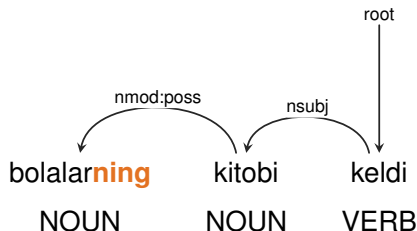
A single suffix can determine the dependency relation:

Morphology breakdown

*bolalar***ning** = “of the children”

bola + lar (PL) + **ning** (GEN)

The **-ning** (GEN) suffix alone drives the `nmod:poss` arc.



“the children’s book arrived”

Example of a manually annotated UzUDT dependency tree.



The gap and our approach

Problem. Existing Uzbek UD treebanks are small; neural parsers need both *more data and better representations*.

New resource

UzUDT — 684 gold UD sentences (largest for Uzbek).

Two lightweight strategies

1. Last-subword **fusion** (suffix-aware).
2. **Cross-treebank** training (merge genres).

No heavy architecture changes — efficient and low-resource-friendly.



Contributions

1. **UzUDT**: a new 684-sentence gold Uzbek UD treebank (INCEpTION, 6 annotators, full adjudication).
2. **Last-subword fusion**: a linguistically-motivated WordPiece→UD-token alignment.
3. **Cross-treebank training**: merging genre-complementary Uzbek treebanks.
4. **Controlled factorial study** (3 configs $\times$ 2 data = 6 runs)
+ released code and 12 checkpoints.

Headline result

Best system **85.08 UPOS / 72.39 UAS / 63.81 LAS**;
cross-treebank adds +9.62 **to** +11.16 **LAS**.



Related work

- ▶ **Universal Dependencies**: cross-lingual syntactic annotation (de Marneffe et al., 2021).
- ▶ **Agglutinative parsing**: morphology-aware features help Turkish / Turkic (Eryiğit et al., 2008).
- ▶ **Uzbek UD**: UD_Uzbek-UT (~500 sents); we add **UzUDT** (684).
- ▶ **Neural pipelines**: Stanza; biaffine parser (Dozat & Manning, 2017); UDify (75 languages).
- ▶ **Uzbek encoders**: TahrirchiBERT (broad domain), BERTbek (news).

Our niche

Lightweight robustness for low-resource Uzbek: *suffix-aware fusion + cross-treebank data.*



UzUDT: a new gold Uzbek UD treebank

- ▶ 684 sentences, ~ 7.8 k tokens.
- ▶ Genres: fiction + academic.
- ▶ INCEpTION; 6 annotators (4 linguists + 2 NLP engineers).
- ▶ Double annotation + full adjudication.
- ▶ UPOS, XPOS, lemma, features, deprels (UD v2).

Inter-annotator agreement

Layer	κ	α
Lemma	95.2	94.6
UPOS	93.7	94.1
Features	90.8	89.9

Currently the **largest publicly available Uzbek UD treebank**.



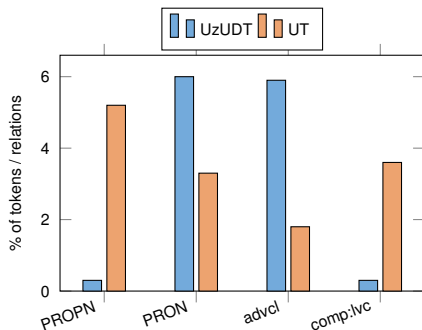
Second treebank & data settings

UD_Uzbek-UT (Akhundjanova & Talamo, 2025): 500 sentences, news + fiction, semi-automatic with manual correction.

Setting	Source	Train	Dev	Test	Total
. 1 UzUDT only	UzUDT	451	45	188	684
. 2 Merged	UzUDT+UT	781	78	325	1,184

Merging nearly **doubles** the training data (451 $\rightarrow$ 781 sentences) — a direct test of whether data quantity is the bottleneck.

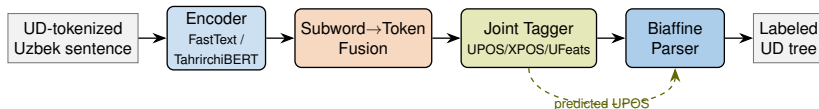
Why the treebanks are complementary



- ▶ UT: many **PROPN** (news; 5.2% vs 0.3%).
- ▶ UzUDT: more **PRON**; 4× more **advcl**.
- ▶ UT: 10× more **compound:lvc**.

⇒ genre-complementary, so merging fills each other's syntactic gaps.

Pipeline: from text to a labeled tree



Encoder is **frozen**; fusion maps WordPieces to UD tokens; training is **sequential** (tagger first, then parser on *re-tagged* data — no gold-tag leakage).

Token representation: four channels

Each UD token is represented by a concatenation of four vectors:

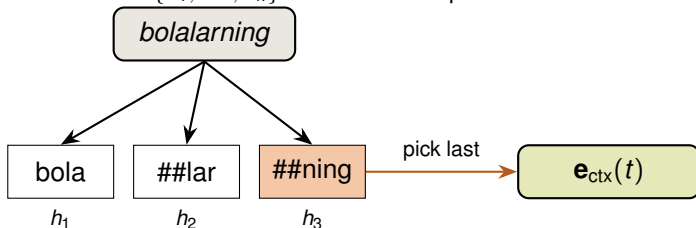
$$\mathbf{x}_i = \left[\underbrace{\mathbf{e}_{\text{word}}(w_i)}_{\text{static}}; \underbrace{\mathbf{e}_{\text{char}}(w_i)}_{\text{char-LSTM}}; \underbrace{\mathbf{e}_{\text{upos}}(\hat{p}_i)}_{\text{predicted}}; \underbrace{\mathbf{e}_{\text{ctx}}(w_i)}_{\text{fused BERT}} \right]$$



- **FastText config:** word channel is static 300-d; *no* ctx channel.
- **TahrirchiBERT config:** ctx channel is a 768-d fused vector (next slide).

Subword→token fusion: our key heuristic

Problem: BERT WordPieces $\neq$ UD tokens — we must aggregate $\{h_1, \dots, h_k\}$ into one vector per token.



Last-subword (ours)

$$e_{\text{ctx}}(t) = h_k$$

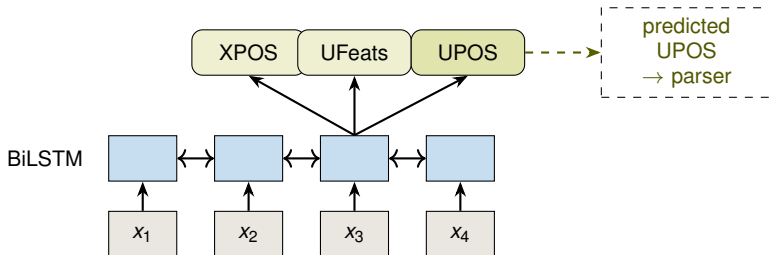
Rightmost piece = outermost suffix (##ning = GEN; ##siz = 2PL).

Mean pooling (baseline)

$$e_{\text{ctx}}(t) = \frac{1}{k} \sum_{i=1}^k h_i$$

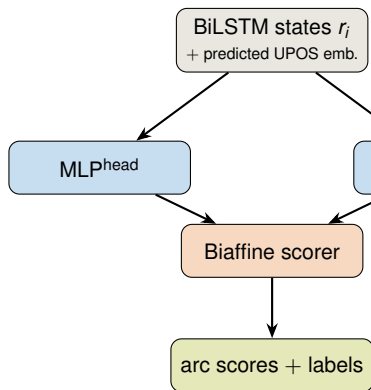
Language-agnostic; dilutes the suffix signal.

Joint morphosyntactic tagger



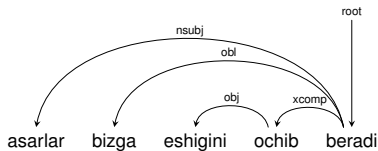
A shared **BiLSTM** feeds three softmax heads. Only the **predicted** UPOS embedding is exported to the parser.

Deep biaffine dependency parser



Dozat & Manning (2017).

- BiLSTM states $\rightarrow$ head/dep subspaces via MLPs.
- Biaffine attention scores every head–dependent pair.
- Output: a labeled tree over UD tokens.



Experiment matrix: 3 configs $\times$ 2 data settings

Run	Embeddings	Fusion	Data	Train
E1.1	FastText	—	UzUDT	451
E1.2	FastText	—	UzUDT+UT	781
E2.1	TahrirchiBERT	Last-subword	UzUDT	451
E2.2	TahrirchiBERT	Last-subword	UzUDT+UT	781
E3.1	TahrirchiBERT	Mean pooling	UzUDT	451
E3.2	TahrirchiBERT	Mean pooling	UzUDT+UT	781

- **E1 vs E2:** embedding effect (static vs contextual).
- **E2 vs E3:** fusion effect (last-subword vs mean).
- **.1 vs .2:** data effect (cross-treebank augmentation).



Main results (test set)

Tagging (accuracy %)

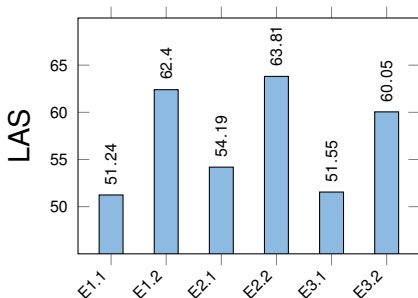
Run	UPOS	XPOS	UFeats
E1.1	79.19	79.81	66.61
E1.2	80.26	83.20	66.98
E2.1	82.45	80.90	65.37
E2.2	85.08	84.72	71.09
E3.1	82.76	81.37	65.22
E3.2	84.02	87.07	70.39

Parsing (accuracy %)

Run	UAS	LAS
E1.1	69.57	51.24
E1.2	72.27	62.40
E2.1	72.05	54.19
E2.2	72.39	63.81
E3.1	69.10	51.55
E3.2	70.74	60.05

E2.2 (BERT + last-subword + merged) wins 4 of 5 metrics; only XPOS is best under mean pooling (E3.2).

Three findings



1. Contextual > static.

BERT vs FastText: UPOS +4.82 on merged data.

2. Fusion matters for parsing.

Last-subword vs mean: LAS +3.76.
mean-BERT 60.05 < FastText 62.40 < last-sub BERT 63.81.

3. Cross-treebank is the biggest lever.

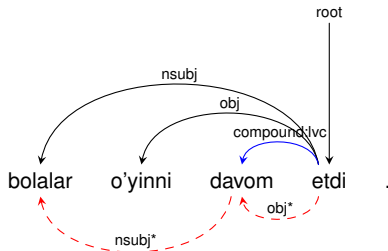
LAS +9.62 (BERT), +11.16 (FastText).

Error analysis (100 test sentences, E2.2)

Error type	%
PP/oblique attach.	28
Coordination scope	22
Subordinate clauses	18
Compound constructions	14
Long-distance deps	10
Other	8

Top sources: *obl/nmod* ambiguity and coordination scope.

Missed light-verb construction:



Blue = gold *compound:lvc*; red dashed = predicted error.



Conclusion

- ▶ **UzUDT**: largest gold Uzbek UD treebank (684 sentences).
- ▶ **Cross-treebank** training is the dominant factor (LAS +9.62 to +11.16).
- ▶ **Last-subword fusion** is critical for parsing (LAS +3.76 over mean pooling).
- ▶ Best: TahrirchiBERT + last-subword + merged = **85.08 / 72.39 / 63.81** (UPOS / UAS / LAS).

Limitations: small dev sets (45 / 78); UT genre mismatch and validation noise.

Future: BERTbek, BERT+FastText fusion (E4–E7).

Data & 12 checkpoints: <https://huggingface.co/Sanatbek/uzudt>

Code: <https://github.com/SanatbekMatlatipov/robust-parsing-uzbek>

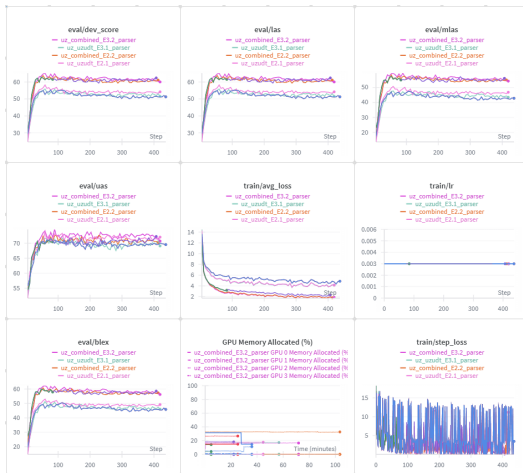


Thank you!

Questions?

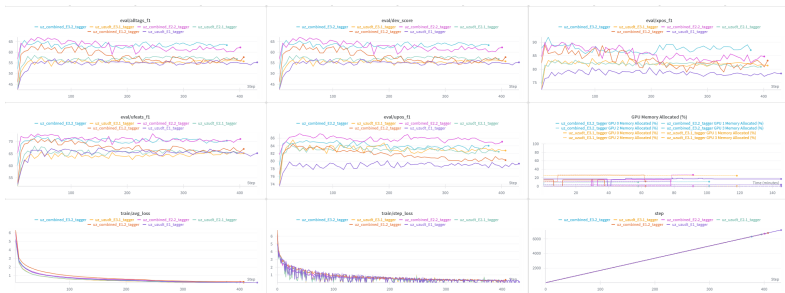
Sanatbek Matlatipov — `s.matlatipov@nuu.uz`
National University of Uzbekistan named after Mirzo Ulugbek

Appendix: parser training dynamics



Dev score and LAS/UAS/MLAS/BLEX over training steps (UzUDT-only vs. merged).

Appendix: tagger training dynamics



UPOS/XPOS/UFeats/AllTags F1 across the six runs.

Appendix: a correctly parsed sentence

